

Multiparameter Persistent Homology vs. Classical Baselines

Concise Summary — Thomas and Nested-Thomas Point Processes

point-process-tda

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1 Setup

- 4 methods, 2 processes, 3 seeds each (9371–9373), identical train/val/test split per seed within a process.
- **vihrs**: 1-D CNN over Ripley $L(r) - r + n(x)$ (Vihrs 2022 baseline).
- **pi_multik (k=5,10,15)**: shared CoordConv encoder over DTM persistence images at 3 density scales, late-fused.
- **mph_pi**: CoordConv encoder over one joint (density, radius) bifiltration image — same architecture as **pi_multik**’s branch, no k -axis (scale is already a filtration axis).
- **mph_fusion**: **vihrs** branch + **mph_pi** branch, late-fused.
- All targets: log-transformed, then z -scored (train-split mean/std). Loss = MSE on that standardized scale, always.

2 Reading the loss numbers

- Loss is standardized log-MSE, not raw physical error — 0.03 does not mean the same thing for every target.
- No-skill baseline ≈ 1.0 (predicting the training mean). $R^2 \approx 1 - \text{loss}$.
- To physical units: $\text{RMSE}_z = \sqrt{\text{loss}}$; typical prediction is within a factor of $\exp(\text{RMSE}_z \cdot \sigma_{\log})$ of the truth, where $\sigma_{\log}$ is that target’s log-space std (varies by target — Table 1).
- Example: loss 0.03 on **cluster_scale** ($\sigma_{\log} = 0.655$) $\rightarrow R^2 = 0.97$, $\pm 12\%$ typical error. Same loss on **meta_offspring** ($\sigma_{\log} = 0.409$) $\rightarrow$ same R^2 , tighter $\pm 7\%$.
- Compare loss freely *within* a target/process; always translate before comparing *across* targets.

Loss	R^2	parent_intensity	meta_offspring	meta_cluster_scale	mean_offspring	cluster_scale
0.50	0.50	42%	34%	56%	44%	59%
0.20	0.80	25%	20%	32%	26%	34%
0.10	0.90	17%	14%	22%	18%	23%
0.05	0.95	12%	10%	15%	12%	16%
0.03	0.97	9%	7%	11%	9%	12%
0.01	0.99	5%	4%	6%	5%	7%

Table 1: Nested Thomas: standardized loss $\rightarrow R^2 \rightarrow$ typical multiplicative error per target (e.g. ”9%” means a typical prediction is within $\pm 9\%$ of the true value).

Loss	R^2	parent_intensity	mean_offspring	cluster_scale
0.50	0.50	50%	63%	63%
0.20	0.80	29%	36%	36%
0.10	0.90	20%	24%	24%
0.05	0.95	14%	17%	17%
0.03	0.97	11%	13%	13%
0.01	0.99	6%	7%	7%

Table 2: Thomas: standardized loss $\rightarrow R^2 \rightarrow$ typical multiplicative error per target (e.g. "9%" means a typical prediction is within $\pm 9\%$ of the true value).

3 Nested Thomas

- Two-level cluster process, 5 targets, 8000 clouds (7000 train/val/test + 1000 adversarial).

Method	Test loss [†]	Adversarial loss [†]
vihrs	0.2171 ± 0.0037	0.2194 ± 0.0055
pi_multik	0.1927 ± 0.0029	0.2000 ± 0.0017
mph_pi	0.2095 ± 0.0045	0.2186 ± 0.0031
mph_fusion	0.1965 ± 0.0022	0.1966 ± 0.0009

Table 3: Nested Thomas: overall test/adversarial loss, mean $\pm$ std over 3 seeds. [†]standardized log-MSE (Section 2), not raw physical units – see Table 1 to translate.

Method	parent_intensity	meta_offspring	meta_cluster_scale	mean_offspring	cluster_scale
vihrs	0.320	0.494	0.132	0.112	0.028
pi_multik	0.286	0.410	0.142	0.086	0.040
mph_pi	0.289	0.449	0.158	0.109	0.042
mph_fusion	0.280	0.445	0.131	0.097	0.029

Table 4: Nested Thomas: per-target test loss (standardized log-MSE, Section 2), mean over 3 seeds – see Table 1 to translate, corresponding figure for error bars.

Findings:

- Overall test loss: **pi_multik** (0.1927) $\approx$ **mph_fusion** (0.1965) < **mph_pi** (0.2095) < **vihrs** (0.2171).
- No method wins every target.
- **mph_pi** > **vihrs** on intensity/count targets (**parent_intensity**, **meta_offspring**).
- **vihrs** > **mph_pi** on **cluster_scale** — $L(r) - r$ targets clustering radius directly; the bifiltration image is a density/topology signal.
- **mph_fusion** wins **parent_intensity** and **meta_cluster_scale** outright; beats both parent methods overall in 3/3 seeds.
- **pi_multik** wins **meta_offspring** and **mean_offspring** outright — 3 fixed DTM scales capture nearly what the joint bifiltration does here.
- **vihrs** alone wins only **cluster_scale**, but wins it clearly.

nested thomas: training curves (seed 9371)

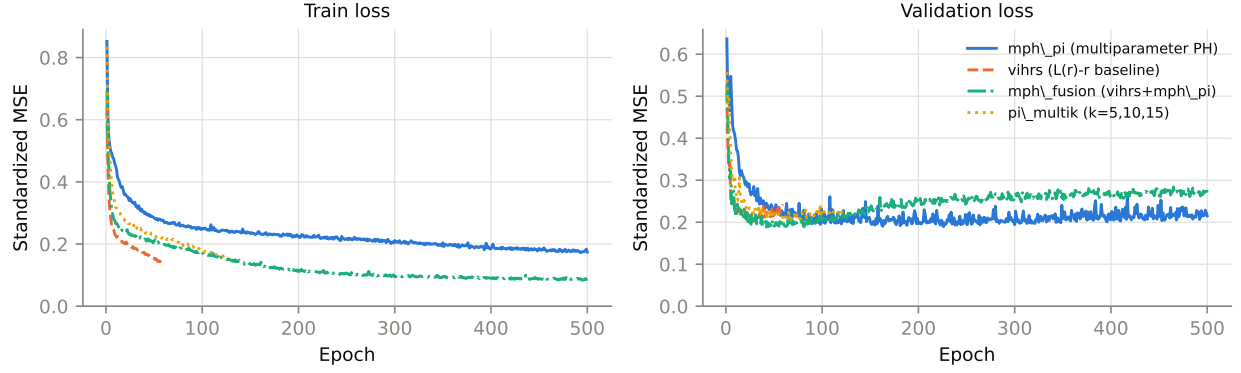


Figure 1: Loss curves, seed 9371. All 4 methods reload the best-val-loss checkpoint before final eval regardless of stopping rule.

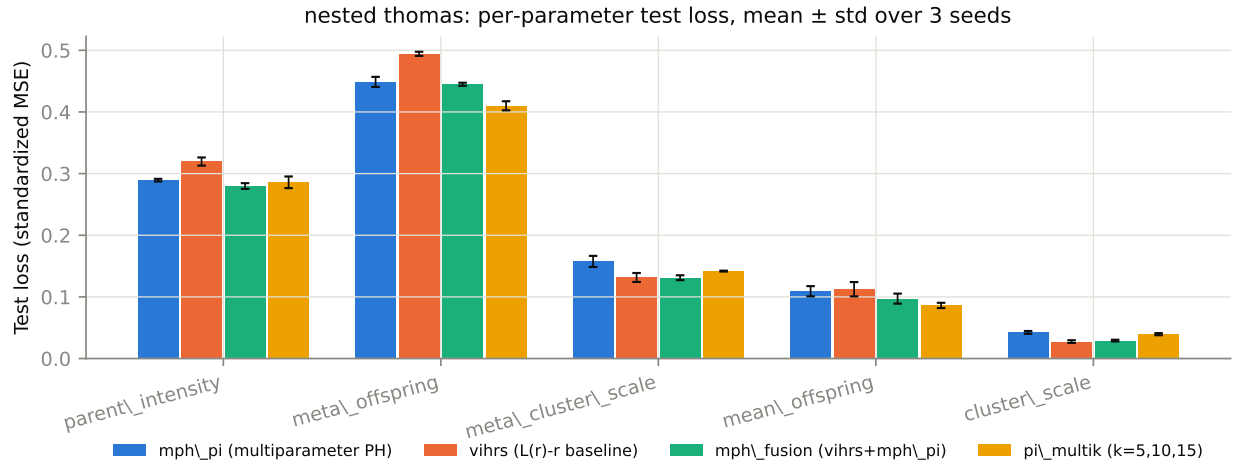


Figure 2: Per-target test loss, mean $\pm$ std over 3 seeds.

- Takeaway: representation choice trades off *which* targets it resolves best; fusion is the more robust default.

4 Thomas

- Single-level cluster process, 3 targets, same 8000-cloud split convention.
- `pi_multik` (and, after a local crash, `vihrs/mph_pi/ mph_fusion` too) trained on HPC and downloaded rather than run locally — see Section 6.

Method	Test loss [†]	Adversarial loss [†]
<code>vihrs</code>	0.1154 ± 0.0091	0.1180 ± 0.0008
<code>pi_multik</code>	0.1248 ± 0.0064	0.1233 ± 0.0027
<code>mph_pi</code>	0.1307 ± 0.0051	0.1289 ± 0.0037
<code>mph_fusion</code>	0.1131 ± 0.0097	0.1148 ± 0.0024

Table 5: Thomas: overall test/adversarial loss, mean $\pm$ std over 3 seeds. [†]standardized log-MSE (Section 2), not raw physical units – see Table 2 to translate.

Method	parent_intensity	mean_offspring	cluster_scale
<code>vihrs</code>	0.193	0.125	0.029
<code>pi_multik</code>	0.202	0.131	0.041
<code>mph_pi</code>	0.210	0.143	0.040
<code>mph_fusion</code>	0.188	0.123	0.029

Table 6: Thomas: per-target test loss (standardized log-MSE, Section 2), mean over 3 seeds – see Table 2 to translate, corresponding figure for error bars.

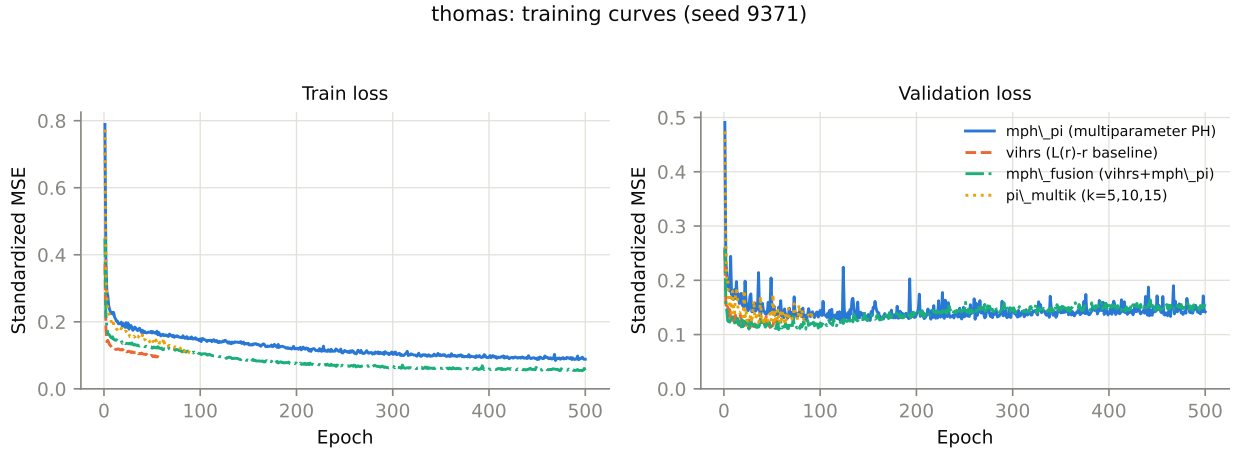


Figure 3: Loss curves, seed 9371.

Findings:

- Overall test loss: `mph_fusion` (0.1131) < `vihrs` (0.1154) < `pi_multik` (0.1248) < `mph_pi` (0.1307).

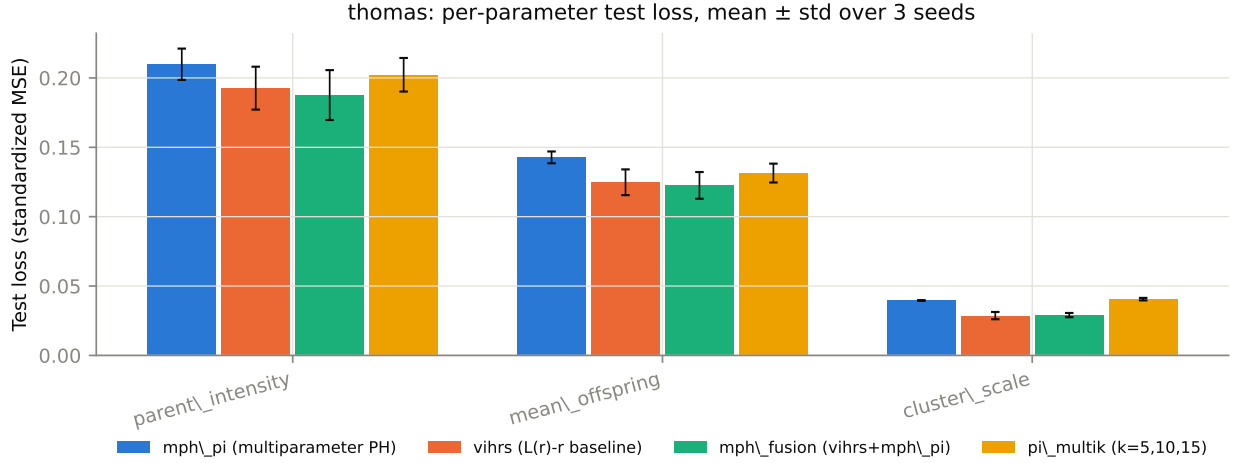


Figure 4: Per-target test loss, mean $\pm$ std over 3 seeds.

- Unlike nested Thomas, `pi_multik` does not lead here — one clustering scale to resolve instead of two, so sweeping 3 DTM k values adds less on top of a single bifiltration or a direct radius estimate.
- `mph_fusion` wins `parent_intensity` and `mean_offspring` outright.
- `vihrs` alone still wins `cluster_scale` clearly (0.029 vs. 0.040 for the DTM-based methods) — same pattern as nested Thomas.
- Takeaway: fusing $L(r) - r$ in rather than discarding it is what wins overall, both processes.

5 Overall

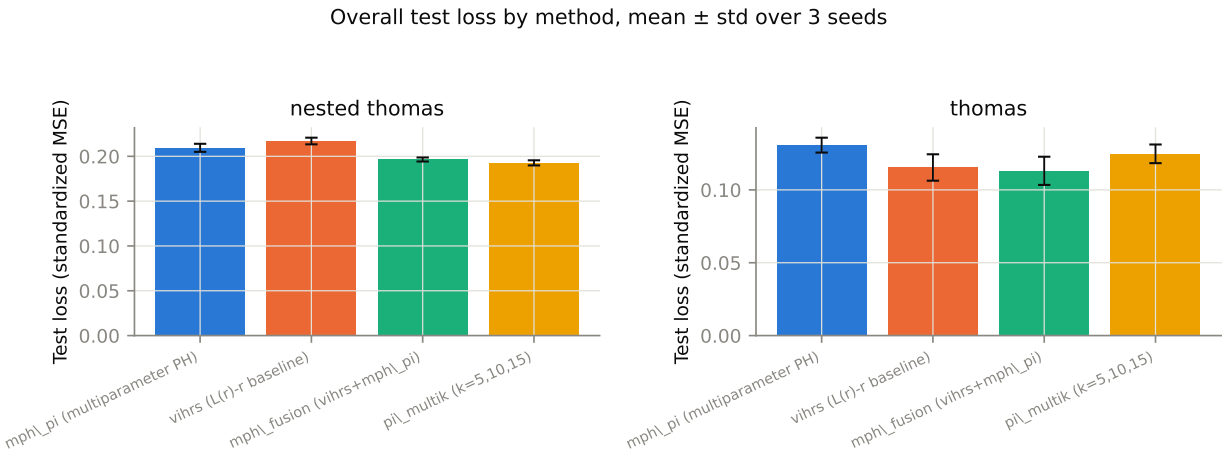


Figure 5: Overall test loss by method, mean $\pm$ std over 3 seeds.

6 Comparability notes

- `vihrs/pi_multik`: bespoke loops, early stopping. `mph_pi/mp_hfusion`: shared runner, fixed epoch budget, no early stopping. All four reload best-val-loss checkpoint before final eval — comparable despite different stopping rules; only wall-clock cost differs.
- Two data issues found and fixed while producing these results: `nested_thomas`'s $k = 5$ diagrams had never finished computing; `thomas`'s point clouds had been lost and were regenerated from the process's seeded design.
- `thomas`: all 4 methods' results were trained on HPC and downloaded (local runs were lost to a crash / proved too memory-heavy to finish reliably); validated on arrival (clean unpickle, correct labels, expected record counts) but not independently re-run locally to confirm byte-identical reproduction.
- Adversarial loss: held-out population, disjoint seed offset, never seen during training/model selection.